

Tarea sección 6.2

$$5) \sec \beta = 2$$

$$\text{Cuad. I} = \frac{\pi}{3} + 2\pi n$$

$$\text{Cuad. IV} = \frac{5\pi}{3} + 2\pi n$$

$$7) \sin x = \frac{\pi}{2}$$

~~No tiene solución, ya que está fuera del dominio~~

$$9) \cos \theta = \frac{1}{\sec \theta}$$

$$\cos \theta = \frac{1}{\frac{1}{\cos \theta}}$$

$$\cos \theta = \cos \theta$$

$$11) 2 \cos 2\theta - \sqrt{3} = 0$$

$$2 \cos 2\theta = \sqrt{3}$$

$$\cos 2\theta = \frac{\sqrt{3}}{2}$$

$$2\theta = \cos^{-1} \frac{\sqrt{3}}{2}$$

$$\theta = \frac{\pi}{12}$$

$$\cos(2\pi - 2\theta) = \frac{\sqrt{3}}{2}$$

$$2\pi - 2\theta = \cos^{-1} \frac{\sqrt{3}}{2}$$

$$\theta_2 = \frac{11\pi}{12}$$

$$15) \sin\left(\theta + \frac{\pi}{4}\right) = \frac{1}{2}$$

$$\theta + \frac{\pi}{4} = \sin^{-1}\left(\frac{1}{2}\right)$$

$$\theta + \frac{\pi}{4} = \frac{\pi}{6}$$

$$\theta_1 = -\frac{\pi}{12} + 2\pi n$$

$$\theta_2 = \frac{7\pi}{12} + 2\pi n$$

$$19) 2\cos t + 1 = 0$$

$$2\cos t = -1$$

$$\cos t = -\frac{1}{2}$$

$$\theta_1 = \frac{2\pi}{3} + 2\pi n$$

$$\theta_2 = \frac{4\pi}{3} + 2\pi n$$

$$23) (\cos \theta - 1) \sin \theta = 0$$

$$\cos \theta - 1 = 0$$

$$\cos \theta = 1$$

$$\sin \theta = 0$$

$$\theta = \pi n$$

* fuera del dominio

$$25) \tan^2 x = 1$$

$$\tan x = \pm \sqrt{1}$$

$$\tan x = 1$$

$$\tan x = -1$$

$$x = \frac{\pi}{4} + 2\pi n$$

$$31) (2 \operatorname{sen} \theta + 1)(2 \cos \theta + 3) = 0$$

$$2 \operatorname{sen} \theta + 1 = 0$$

$$2 \operatorname{sen} \theta = -1$$

$$\operatorname{sen} \theta = \frac{-1}{2}$$

$$2 \cos \theta + 3 = 0$$

$$2 \cos \theta = -3$$

$$\cos \theta = \frac{-3}{2}$$

$$\theta_1 = \frac{7\pi}{6} + 2\pi n$$

$$\theta_2 = \frac{11\pi}{6} + 2\pi n$$

$$35) \operatorname{sen} 2x (\csc 2x - 2) = 0$$

$$\operatorname{sen} 2x = 0$$

$$\csc 2x - 2 = 0$$

$$\csc 2x = 2$$

* fuera del dominio

$$\theta_1 = \frac{\pi}{12} + 2\pi n$$

$$\theta_2 = \frac{5\pi}{12} + 2\pi n$$

$$41) \log(\cos x) = 0$$

$$\cos x = 0$$

$$x = 2\pi n$$

$$49) 2 \operatorname{sen}^2 u = 1 - \operatorname{sen} u$$

$$2 \operatorname{sen}^2 u + \operatorname{sen} u - 1 = 0$$

$$\operatorname{sen} u = \frac{-4}{4}$$

$$\operatorname{sen} u = 1$$

$$\operatorname{sen} u = \frac{-1 \pm \sqrt{1 - 4(-1)(2)}}{4}$$

$$\operatorname{sen} u = \frac{-2}{4}$$

$$\operatorname{sen} u = \frac{1}{2}$$

$$u = \frac{3\pi}{2} + 2\pi n$$

$$u = \frac{\pi}{2} + 2\pi n$$

$$u = \frac{5\pi}{6} + 2\pi n$$

$$u = \frac{\pi}{6} + 2\pi n$$

$$55) 2 \cos^2 y + \cos y = 0$$

$$\cos y (2 \cos y + 1) = 0$$

$$\cos y = 0$$

$$y = \cos^{-1} 0$$

$$y = \frac{\pi}{2}$$

$$y = \frac{3\pi}{2}$$

$$2 \cos y + 1 = 0$$

$$2 \cos y = -1$$

$$\cos y = -\frac{1}{2}$$

$$y = \cos^{-1}(-\frac{1}{2})$$

$$y = \frac{2\pi}{3}$$

$$y = \frac{4\pi}{3}$$

$$63) 2 \tan t - \sec^2 t = 0$$

$$2 \frac{\sin t}{\cos t} - \left(\frac{1}{\cos t} \right)^2 = 0$$

$$\frac{2 \sin t}{\cos t} - \frac{1}{\cos^2 t} = 0$$

$$\frac{2 \cos t \sin t - 1}{\cos^2 t} = 0$$

$$2 \cos t \sin t - 1 = 0$$

$$\sin 2t - 1 = 0$$

$$\sin 2t = 1$$

$$2t = \sin^{-1} 1$$

$$2t = \frac{\pi}{2}$$

$$t = \frac{\pi}{4}$$

$$2t = \frac{5\pi}{2}$$

$$t = \frac{5\pi}{4}$$

$$2t = \frac{9\pi}{2}$$

$$t = \frac{9\pi}{4}$$

$$(65) \cot x + \tan x = \csc x \sec x$$

$$\frac{\cos x}{\sin x} + \frac{\sin x}{\cos x} = \frac{1}{\sin x} \left(\frac{1}{\cos x} \right)$$

$$\frac{\cos x}{\sin x} + \frac{\sin x}{\cos x} = \frac{1}{\sin x \cos x}$$

$$\frac{\cos x}{\sin x} + \frac{\sin x}{\cos x} - \frac{1}{\sin x \cos x} = 0$$

$$\frac{\cos^2 x + \sin^2 x - 1}{\sin x \cos x} = 0$$

$$\cos^2 x + \sin^2 x - 1 = 0$$

$$1 - 1 = 0$$

$$\underline{0 = 0}$$